IP Addressing And Subnetting

1. List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.

Ans:-

Class	First octet range	Default mask	Used for	Real-world example
A	1 - 126	255.0.0.0 (/8)	Very large networks	Large company or cloud network using the private range 10.x.x.x
B	128 - 191	255.255.0.0 (/16)	Medium to large networks	College or university campus network (e.g. 172.16.x.x)
C	192 - 223	255.255.255.0 (/24)	Small networks	Home WiFi router (192.168.1.1) with phones and laptops
D	224 - 239	Not used (multicast)	Sending data to many devices at once	IPTV or live video streaming to many viewers
E	240 - 255	Not used (reserved)	Experimental / research	Reserved for testing, not used by normal devices

Note: 127.x.x.x is reserved for loopback (testing your own device), so Class A usable range is 1 - 126.

2. Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses.

Hint: Show your working steps for each calculation.

   Ans:-

Step 1 (Network address): Network address = IP address AND subnet mask. The mask 255.255.255.0 Keeps the first three octets and makes the last octet 0, so 192.168.10.45 becomes 192.168.10.0.

Step 2 (Broadcast address): Set all host bits (last octet, 8 bits) to 1. The last octet becomes 11111111 = 255, so the broadcast address is 192.168.10.255.

Step 3 (Valid hosts): First host = network address + 1 = 192.168.10.1. Last host = broadcast address - 1 = 192.168.10.254. Number of hosts = 2^8 - 2 = 254.

Step	Octet 1	Octet 2	Octet 3	Octet 4	Result
IP address (binary)	11000000	10101000	00001010	00101101	192.168.10.45
Subnet mask (binary)	11111111	11111111	11111111	00000000	255.255.255.0
Network = IP AND mask	11000000	10101000	00001010	00000000	192.168.10.0
Broadcast = host bits all 1	11000000	10101000	00001010	11111111	192.168.10.255

Network address: 192.168.10.0

Broadcast address: 192.168.10.255

Valid host range: 192.168.10.1 to 192.168.10.254 (254 hosts)

3. Create a Subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.

Ans:-

Step 1: We need 4 subnets. 2^2 = 4, so we borrow 2 bits from the host part.

Step 2: New prefix = /24 + 2 = /26. The subnet mask is 255.255.255.192 (last octet 11000000 = 192).

Step 3: Block size = 256 - 192 = 64, so the subnets start at 0, 64, 128 and 192.

Step 4: Hosts per subnet = 2^6 - 2 = 62.

Group	Network address	Subnet mask	Valid host range	Broadcast address
Delivery Partners	10.0.0.0/26	255.255.255.192	10.0.0.1 - 10.0.0.62	10.0.0.63
Restaurants	10.0.0.64/26	255.255.255.192	10.0.0.65 - 10.0.0.126	10.0.0.127
Customers	10.0.0.128/26	255.255.255.192	10.0.0.129 - 10.0.0.190	10.0.0.191
Admins	10.0.0.192/26	255.255.255.192	10.0.0.193 - 10.0.0.254	10.0.0.255

Each subnet has 62 usable host addresses.

4. You are given the following scenario: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet.

Constraint: Show your calculation steps and final answers.

   Ans:-

Step 1: We need at least 20 subnets. 2^4 = 16 is too small and 2^5 = 32 is enough, so we borrow 5 bits.

Step 2: New prefix = /16 + 5 = /21.

Step 3: Subnet mask: the first 21 bits are 1s, so it is 11111111.11111111.11111000.00000000 = 255.255.248.0.

Step 4: Host bits = 32 - 21 = 11. Hosts per subnet = 2^11 - 2 = 2048 - 2 = 2046.

Step 5: Block size in the third octet = 256 - 248 = 8, so the subnets are 172.16.0.0, 172.16.8.0, 172.16.16.0 and so on.

Subnet no.	Network address	Valid host range	Broadcast address
1	172.16.0.0/21	172.16.0.1 - 172.16.7.254	172.16.7.255
2	172.16.8.0/21	172.16.8.1 - 172.16.15.254	172.16.15.255
3	172.16.16.0/21	172.16.16.1 - 172.16.23.254	172.16.23.255
...	...	...	...
32	172.16.248.0/21	172.16.248.1 - 172.16.255.254	172.16.255.255

New subnet mask: 255.255.248.0 (/21)

Number of subnets: 32

Hosts per subnet: 2046

5. A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.

   Ans:-

Hi, the problem is that 192.168.1.300 is not a valid IPv4 address. Each part (octet) of an IPv4 address uses 8 bits, so it can only be a number from 0 to 255. Your last number is 300, which is too big, so the network cannot accept it.

In the 192.168.1.x network (mask 255.255.255.0), a valid address must have a last number from 1 to 254, for example 192.168.1.30 or 192.168.1.100. Do not use 192.168.1.0 (network address) or 192.168.1.255 (broadcast address). The easiest fix is to set your device to get the IP automatically (DHCP), or type a valid address like the ones above.